

# GA1025 Macro Theory

## Solution to Problem Set 3

Junbiao Chen\*

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### Problem 1.1

The sequence problem of the Cass-Koopmans model is given by

$$\begin{aligned} V^*(k_0) &= \max_{\{k_{t+1}\}_0^\infty} \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\gamma} - 1}{1-\gamma} \\ \text{subject to } k_{t+1} &= (1-\delta)k_t + Ak_t^\alpha - c_t \\ k_0 &\text{ is given} \end{aligned}$$

After substituting the resource constraints, we have

$$V^*(k_0) = \max_{\{k_{t+1}\}_0^\infty} \sum_{t=0}^{\infty} \beta^t \frac{((1-\delta)k_t + Ak_t^\alpha - k_{t+1})^{1-\gamma} - 1}{1-\gamma}$$

Take the derivative w.r.t.  $k_{t+1}$ , we have

$$\beta^t \left[ (1-\delta)k_t + Ak_t^\alpha - k_{t+1} \right]^{-\gamma} (-1) + \beta^{t+1} \left[ (1-\delta)k_{t+1} + Ak_{t+1}^\alpha - k_{t+2} \right]^{-\gamma} \left( 1 - \delta + A\alpha k_{t+1}^{\alpha-1} \right) = 0$$

After simplification, we have

$$\left[ (1-\delta)k_t + Ak_t^\alpha - k_{t+1} \right]^{-\gamma} = \beta \left[ (1-\delta)k_{t+1} + Ak_{t+1}^\alpha - k_{t+2} \right]^{-\gamma} \left( 1 - \delta + A\alpha k_{t+1}^{\alpha-1} \right)$$

In the steady state,  $k_t = k_{t+1} = k_{t+2} = \bar{k}$ , as a result, we can compute the ss  $\bar{k}$  as follows.

$$\bar{k} = \left( \frac{1 - \beta + \delta\beta}{A\alpha\beta} \right)^{\frac{1}{\alpha-1}}$$

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\*E-mail: jc14076@nyu.edu.

After substituting  $c_t$  back to the Euler equation, we have

$$c_t^{-\gamma} = \beta(1 - \delta + A\alpha k_{t+1}^{\alpha-1})c_{t+1}^{-\gamma}$$

The equation above implies the behavior of consumption smoothing, where the agent equalizes the marginal utility of spending today (LHS) and the marginal utility of spending tomorrow (RHS).

### Problem 1.2

The FE formulation of the Cass-Koopmans model is given by

$$V(k) = \max_{k' \in \Gamma(k)} \frac{c^{1-\gamma} - 1}{1 - \gamma} + \beta V(k')$$

subject to  $k' = (1 - \delta)k + Ak^\alpha - c$

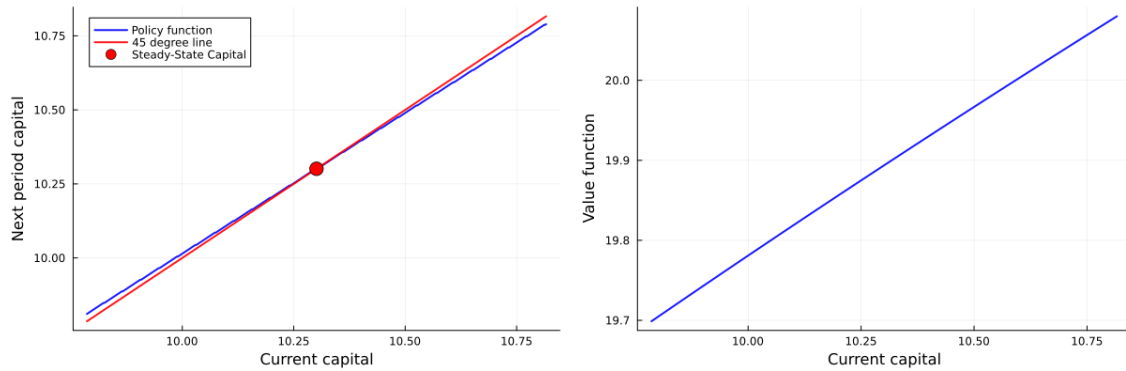
### Problem 1.3

I implemented VFI with the following computational parameters:

| Computational parameters | Values                                                |
|--------------------------|-------------------------------------------------------|
| $k_{min}$                | $0.95 \cdot k_{ss}$                                   |
| $k_{max}$                | $1.05 \cdot k_{ss}$                                   |
| $k_{vec}$                | $k_{min} : \frac{k_{max} - k_{min}}{I - 1} : k_{max}$ |
| max_iter                 | 1000 (maximum number of iterations)                   |
| tol                      | $10^{-8}$ (tolerance for convergence)                 |

**Table 0.1:** Summary of Parameters

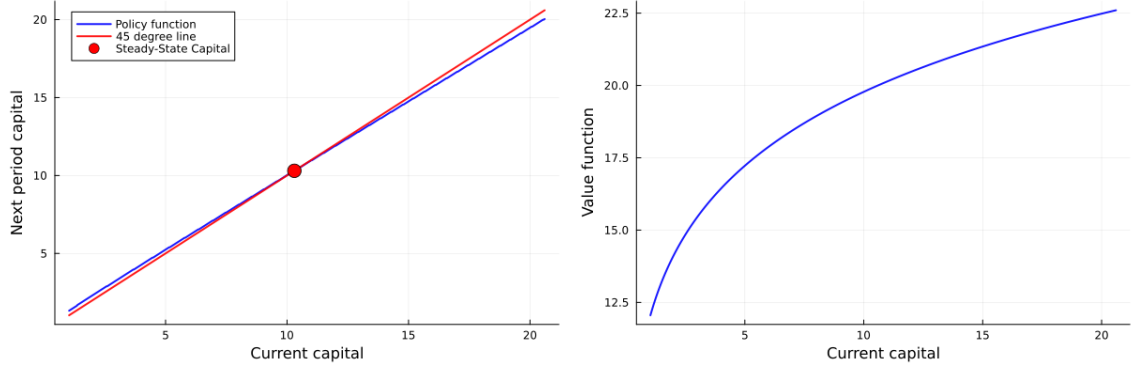
The figures below display the policy function and value function.



**Figure 0.1:** Value Function Iteration

## Problem 1.4

After adjusting the grid, I obtain the same results as 1.3 (shown as follows).



**Figure 0.2:** Value Function Iteration

## Shooting Algorithm

### Problem 1.10

The objective function for the sequence problem in the adjustment model is given by

$$V^*(x_0) = \max_{\{x_{t+1}\}_0^\infty \in \mathbb{R}} \sum_{t=0}^{\infty} \beta^t \left[ -\frac{\phi}{2} x_{t+1}^2 - \frac{\kappa}{2} (x_{t+1} - x_t)^2 \right]$$

Take the first-order condition with respect to  $x_{t+1}$ , we have

$$\beta^t [-\phi x_{t+1} - \kappa(x_{t+1} - x_t)] + \beta^{t+1} \kappa(x_{t+2} - x_{t+1}) = 0$$

After simplification, we have

$$\beta \kappa (x_{t+2} - x_{t+1}) = \phi x_{t+1} + \kappa (x_{t+1} - x_t) \quad (0.1)$$

According to the Definition, the transversality condition is given by

$$\lim_{t \rightarrow \infty} \beta^t \kappa (x_{t+1} - x_t) x_t = 0$$

### Answer to Question 1.11

From equation (0.1), the characteristic function is given by

$$m^2 - \left( \frac{\phi + \kappa}{\beta \kappa} + 1 \right) m + \frac{1}{\beta} = 0$$

It follows that the roots are

$$m_{1,2} = \frac{\left(\frac{\phi+\kappa}{\beta\kappa} + 1\right) \pm \sqrt{\left(\frac{\phi+\kappa}{\beta\kappa} + 1\right)^2 - \frac{4}{\beta}}}{2}$$

As a result,  $m_1 \cdot m_2 = \frac{1}{\beta}$ .

### Answer to Question 1.12

Therefore, the general solution to the problem is given by

$$x_t = A \left( \frac{\left(\frac{\phi+\kappa}{\beta\kappa} + 1\right) - \sqrt{\left(\frac{\phi+\kappa}{\beta\kappa} + 1\right)^2 - \frac{4}{\beta}}}{2} \right)^t + B \left( \frac{\left(\frac{\phi+\kappa}{\beta\kappa} + 1\right) + \sqrt{\left(\frac{\phi+\kappa}{\beta\kappa} + 1\right)^2 - \frac{4}{\beta}}}{2} \right)^t$$

Let  $m_1 < m_2$ , since  $m_1 \cdot m_2 = \frac{1}{\beta}$ , we have

$$\begin{aligned} m_1 &< \frac{1}{\beta m_1} \\ \Rightarrow m_1 &< \beta^{-1/2} \quad \text{and} \quad m_2 > \beta^{-1/2} \end{aligned}$$

Then, we can express  $m_1$  and  $m_2$  as

$$\begin{aligned} m_1 &= \beta^{-1/2}\epsilon, \quad \epsilon \in (0, 1) \\ m_2 &= \beta^{-1/2}\delta, \quad \delta > 1 \end{aligned}$$

Substituting  $x_t$  with  $A(\beta^{-1/2}\epsilon)^t + B(\beta^{-1/2}\delta)^t$ , the Transversality condition becomes

$$\begin{aligned} \lim_{t \rightarrow \infty} \beta^t \left[ A(\beta^{-1/2}\epsilon)^{t+1} + B(\beta^{-1/2}\delta)^{t+1} - A(\beta^{-1/2}\epsilon)^t - B(\beta^{-1/2}\delta)^t \right] \cdot \left[ A(\beta^{-1/2}\epsilon)^t + B(\beta^{-1/2}\delta)^t \right] = \\ \lim_{t \rightarrow \infty} \beta^t \left[ A(\beta^{-1/2}\epsilon)^t (\beta^{-1/2}\epsilon - 1) + B(\beta^{-1/2}\delta)^t (\beta^{-1/2}\delta - 1) \right] \cdot \left[ A(\beta^{-1/2}\epsilon)^t + B(\beta^{-1/2}\delta)^t \right] \leq \\ \lim_{t \rightarrow \infty} A^2 \epsilon^{2t} (\beta^{-1/2}\epsilon - 1) + B^2 \delta^{2t} (\beta^{-1/2}\delta - 1) \end{aligned}$$

Because  $\delta > 1$ ,  $\delta^{2t} \rightarrow \infty$ , therefore, this equation converges to zero if  $B = 0$ , where  $B$  is the coefficient of the larger root.

### Answer to Question 1.13

### Answer to Question 1.14

Take the first-order condition with respect to  $\kappa$  for  $x_t$ , we have

$$\frac{\partial x_t}{\partial \kappa} =$$

and

$$\frac{\partial x_t}{\partial \phi} =$$

## Guess and Verify

### Answer to Question 1.15

Incorporating the budget constraint, we can rewrite the model as follows.

$$\max_{k_{t+1} \in \Gamma(k_t)} \sum_{t=0}^{\infty} \beta^t U(G(k_t) + (1 - \delta)k_t - k_{t+1})$$

$$\text{where } \Gamma(k_t) = [0, G(k_t) + (1 - \delta)k_t]$$

Therefore, the Bellman equation is given by

$$V(k) = \max_{k'} U(G(k) + (1 - \delta)k - k') + \beta V(k')$$

### Answer to Question 1.16

If  $U(c) = \log(c)$ ,  $G(k) = k^\alpha$ , and  $\delta = 1$ , and we “guess” the value function is  $V(k) = a + b \log k$ , the Bellman equation formulated above becomes

$$a + b \log k = \max_{k'} \log(k^\alpha - k') + \beta(a + b \log k')$$

Then,

$$[k'] = -\frac{1}{k^\alpha - k'} + \beta \frac{b}{k'} = 0$$

As a result, the policy function is

$$k' = \frac{\beta b k^\alpha}{1 + \beta b}$$

Substituting  $k' = \frac{\beta b k^\alpha}{1 + \beta b}$  back to the Bellman equation, we have

$$a + b \log k = \log \left( k^\alpha - \frac{\beta b k^\alpha}{1 + \beta b} \right) + \beta a + \beta \log \left( \frac{\beta b k^\alpha}{1 + \beta b} \right)$$

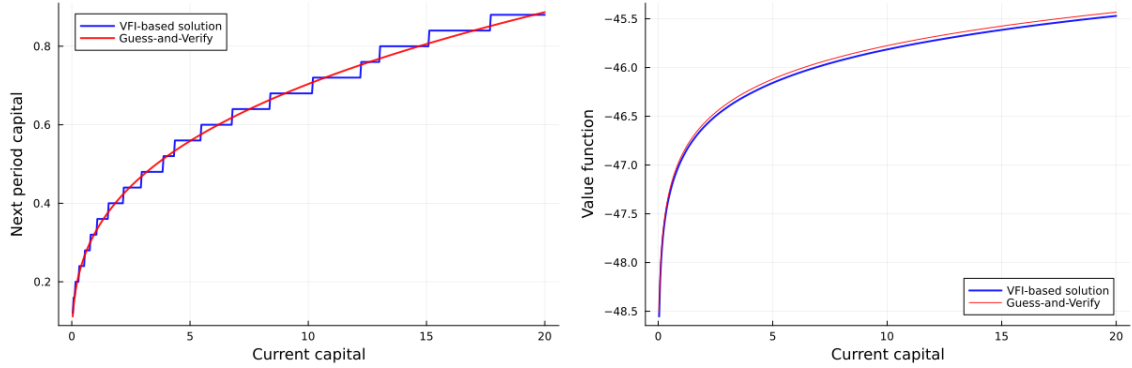
By comparing coefficients, we have

$$b = \frac{\alpha}{1 - \alpha\beta}$$

$$a = \frac{1}{1 - \beta} \left[ \frac{1}{1 - \alpha\beta} \log(1 - \alpha\beta) + \frac{\alpha\beta}{1 - \alpha\beta} \log \frac{\alpha\beta}{1 - \alpha\beta} \right]$$

### Answer to Question 1.17

I verify the analytical solutions above by comparing them to the numerical solutions obtained from the VFI. The figures below display the comparisons.



**Figure 0.3:** Guess-and-Verify

## 2. Probability Theory

### Exercise 2.1

In a stochastic environment of a fair dice, the filtered probability space is given by

- $\Omega := \{1, 2, 3, 4, 5, 6\}^\infty$  denotes the sample space.
- $\mathcal{F} := 2^\Omega$
- $\{\mathcal{F}_t\}_t^\infty$  is defined as follows:

$$\begin{aligned} \mathcal{F}_1 &= 2^{\{1,2,3,4,5,6\}} \\ &\dots \\ \mathcal{F}_t &= 2^{\{1,2,3,4,5,6\}^t} \\ &\dots \\ \mathcal{F}_\infty &= 2^{\{1,2,3,4,5,6\}^\infty} \end{aligned}$$

with  $\mathcal{F}_t \subseteq \mathcal{F}_{t+s}$  for any  $s \geq 0$ . Economically,  $\mathcal{F}_t$  contains the information up to time  $t$ .

- $P = \frac{1}{2^\infty}$

### Exercise 2.2

**Q1.** The sample space is given by  $\Omega = \{H, T\}^T$ . The sample space contains  $2^T$  elements.

**Q2.** The sigma-algebra is given by  $\mathcal{F} = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}^T$ .

**Q3.** Take  $T = 4$ , and  $s^2 = \{H, T\}$ ,  $F(s^t)$  includes the following

1.  $(H, T, H, H)$
2.  $(H, T, H, T)$
3.  $(H, T, T, T)$
4.  $(H, T, T, H)$

**Q4.** Note that for a fixed partial path  $s^t$ , there are  $2^{T-t}$  possibilities. Therefore, the probability measure  $P$  representing fair iid coin flips is given by

$$P(s^T) = \frac{1}{2^T}$$

$$P(F(s^t)) = \frac{2^{T-t}}{2^T} = \frac{1}{2^t}$$

**Q5.** For  $T = 3$ ,

$$\mathcal{F}_0 = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$$

$$\begin{aligned}
\mathcal{F}_1 = \{ & \emptyset, \\
& \{(H, H)\}, \{(H, T)\}, \{(T, H)\}, \{(T, T)\}, \\
& \{(H, H), (H, T)\}, \\
& \{(H, H), (T, H)\}, \\
& \{(H, H), (T, T)\}, \\
& \{(H, T), (T, H)\}, \\
& \{(H, T), (T, T)\}, \\
& \{(T, H), (T, T)\}, \\
& \{(H, H), (H, T), (T, H)\}, \\
& \{(H, H), (H, T), (T, T)\}, \\
& \{(H, H), (T, H), (T, T)\}, \\
& \{(H, T), (T, H), (T, T)\}, \\
& \{(H, H), (H, T), (T, H), (T, T)\} \}
\end{aligned}$$

**Q6.** Consider two paths that share the same partial histories up to time  $t$ . Intuitively, their information overlap with each other. Therefore, we cannot assign probability to only one of them, instead, we need to partition the  $s^\infty$  based on their partial histories  $F_t$ , and assign probabilities according to  $F_t$ .

**Q7.**

1. The smallest  $\sigma$ -algebra  $\mathcal{F}$  is  $\{\emptyset, \Omega, \{s_0, s_1, H, s_3, \dots\}, \{s_0, s_1, T, s_3, \dots\}\}$
2. Denote  $G_t := \{s_0, s_1, \dots : \sum_{i=1}^{10} s_i 1(s_i = H) = t\}$ . Then

$$\mathcal{F} = \bigcup_{t=1}^{10} \{\emptyset, \Omega, 2^{G_t}\}$$

3. Denote  $G'_t := \{s_0, s_1, \dots : \sum_{i < t} s_i 1(s_i = H) = 0\}$ . Then

$$\mathcal{F} = \bigcup_t \{\emptyset, \Omega, 2^{G'_t}\}$$

**Remark:** The answers above does not depend on whether  $T$  is finite or infinite as long as  $T > 10$  for the second question.

**Q8.**

1.  $\mathbb{E}[x] = 1/2$ .  $\mathbb{E}[x|\mathcal{F}_1] = 1/2$ .  $\mathbb{E}[x|\mathcal{F}_2] = x$ .
2. The unconditional and conditional expectations are given by

$$\begin{aligned}\mathbb{E}[x] &= \sum_{t=1}^{10} t \frac{\binom{10}{t}}{2^{10}} \\ \mathbb{E}[x|s_1 = H] &= \sum_{t=1}^{10} t \frac{\binom{9}{t-1}}{2^9} \\ \mathbb{E}[x|s_1 = T] &= \sum_{t=1}^9 t \frac{\binom{9}{t-1}}{2^9} \\ \mathbb{E}[x|s_1, s_2 = H, H] &= \sum_{t=2}^{10} t \frac{\binom{8}{t-2}}{2^8} \\ \mathbb{E}[x|s_1, s_2 = H, T] &= \sum_{t=1}^9 t \frac{\binom{8}{t-1}}{2^8} \\ \mathbb{E}[x|s_1, s_2 = T, H] &= \sum_{t=1}^9 t \frac{\binom{8}{t-1}}{2^8} \\ \mathbb{E}[x|s_1, s_2 = T, T] &= \sum_{t=1}^8 t \frac{\binom{8}{t-1}}{2^8}\end{aligned}$$

3. The unconditional and conditional expectations are given by

$$\begin{aligned}\mathbb{E}[x] &= \sum_{t=1}^{\infty} t \left(\frac{1}{2}\right)^t = 2 \\ \mathbb{E}[x|s_1 = H] &= 0 \\ \mathbb{E}[x|s_1 = T] &= \sum_{t=2}^{\infty} t \left(\frac{1}{2}\right)^{t-1} = 3 \\ \mathbb{E}[x|s_1, s_2 = H, H] &= 0 \\ \mathbb{E}[x|s_1, s_2 = H, T] &= 0 \\ \mathbb{E}[x|s_1, s_2 = T, H] &= 0 \\ \mathbb{E}[x|s_1, s_2 = T, T] &= 4\end{aligned}$$

**Q9.** The Radon-Nikodym derivative of  $\tilde{P}$  with respect to  $P$  for any partial history  $s^t$  is given by

$$\begin{aligned}h(s^t) &= \frac{\sum_{s^{t+j}|s^t} \tilde{P}(s^{t+j})}{P(s^t)} \\ &= \frac{\prod_{i=1}^t p^{(\#\text{heads in } s^i)} (1-p)^{(\#\text{tails in } s^i)}}{(1/2)^t}\end{aligned}$$

**Q10.** Under the fair specification,  $(\mathcal{F}, P)$ , w.l.o.g, consider  $s = t + 1$

$$\mathbb{E}[x_{t+1}|\mathcal{F}_t] = x_t + p + (1 - p)(-1)$$

which is equal to  $x_t$  iff  $p = 1 - p$ . Therefore,  $x_t$  is a martingale with respect to the filtration  $\{\mathcal{F}_t\}_{t \geq 0}$  and  $P$ , but not with respect to  $\{\mathcal{F}_t\}_{t \geq 0}$  and  $\tilde{P}$ ,

**Q11.** The stochastic process in Q10 is time-invariant because

$$\mathbb{E}[x_{t+1}|x] = x + p + (1 - p)(-1)$$

which only depends on  $x$  not  $t$ .

**Q12.** If  $\hat{P}(s_t = H|s_{t-1} = H) = \hat{P}(s_t = T|s_{t-1} = T) = p$ , then

$$\begin{aligned}\mathbb{E}[x_{t+1}|x_t] &= x_t + p + (1 - p)(-1) \text{ if } s_t = H \\ \mathbb{E}[x_{t+1}|x_t] &= x_t + (1 - p) + p(-1) \text{ if } s_t = T\end{aligned}$$

Therefore,  $\mathbb{E}[x_{t+1}|x_t] = x_t$  if  $p = 1/2$ .